

"WEEK-5 GRADED ASSIGNMENT"

Q.1]

Soln:-

$$\sinh(at) \cdot f(t) = \frac{e^{at} - e^{-at}}{2} \cdot f(t)$$

$$= \frac{1}{2} [f(t) \cdot e^{at} - f(t) \cdot e^{-at}]$$

$$\text{Laplace transform} = \frac{1}{2} [F(s-a) - F(s+a)]$$

$$\text{As we know } \mathcal{L} f(t) \cdot e^{at} = F(s-a)$$

Q.2] $f(t) = \sinh(2t) \cdot \sin 3t$

$$= \left(\frac{e^{2t} - e^{-2t}}{2} \right) \sin(3t)$$

$$= \frac{1}{2} [e^{2t} \cdot \sin(3t) - e^{-2t} \cdot \sin(3t)]$$

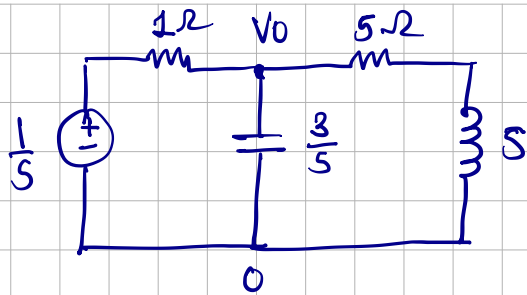
$$= \frac{1}{2} \left[\frac{3}{(s-2)^2 + 9} - \frac{3}{(s+2)^2 + 9} \right]$$

$$= \frac{3}{2} \left[\frac{(s^2 + 2s + 13)}{(s^2 - 2s + 13)} - \frac{(s^2 - 2s + 13)}{(s^2 + 2s + 13)} \right]$$

$$= \frac{3}{2} \left[\frac{4s}{s^4 + 2s^3 + 13s^2 - 2s - 4s^2 - 2s + 13s^2 + 26s + 169} \right]$$

$$= \frac{3}{2} \left[\frac{4s}{s^4 + 22s^2 + 169} \right]$$

Q.3)



$$\frac{s V_0}{3} + V_0 - \frac{1}{s} + \frac{V_0}{5+s} = 0$$

$$\frac{s V_0}{3} + V_0 + \frac{V_0}{5+s} = \frac{1}{s}$$

$$V_0 \left[\frac{s}{3} + \frac{1}{s+5} + 1 \right] = \frac{1}{s}$$

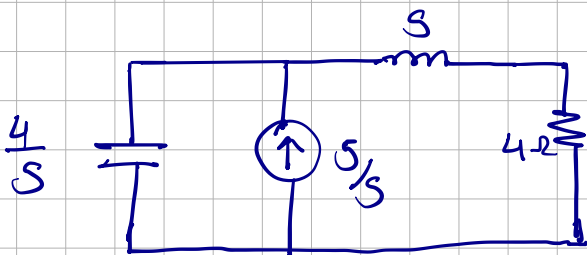
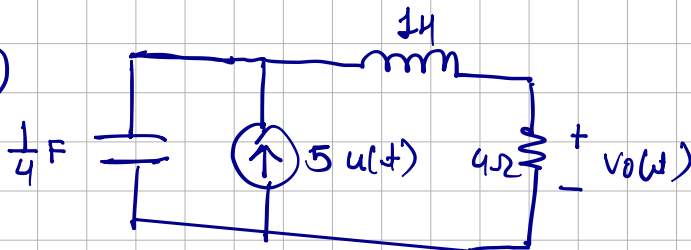
$$V_0 \left[\frac{s(s+5) + 3 + 3(s+5)}{3(s+5)} \right] = \frac{1}{s}$$

$$V_0 \left[\frac{s^2 + 5s + 3 + 3s + 15}{3(s+5)} \right] = \frac{1}{s}$$

$$V_0 \left[\frac{s^2 + 8s + 18}{3(s+5)} \right] = \frac{1}{s}$$

$$V_0 = \frac{3(s+5)}{s(s^2 + 8s + 18)}$$

Q.4)



$$\therefore \frac{\frac{5}{s} \times \frac{4}{s}}{\frac{4}{s} + 4 + s} \times 4 = V_{4\Omega}$$

$$\frac{\frac{20}{s^2}}{s^2 + 4s + 4} \times 4 = V_{4\Omega}$$

$$\frac{80}{s(s+2)^2} = V_4 \Omega$$

$$\frac{1}{s(s+2)^2} = \frac{A}{s} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2}$$

$$\frac{1}{4} = A \quad \text{--- (I)}$$

$$\frac{1}{-2} = C \quad \text{--- (II)}$$

$$\frac{1}{9} = \frac{1}{4} + \frac{B}{3} - \frac{1}{2 \cdot 9}$$

$$\frac{1}{9} \left[1 + \frac{1}{2} \right] = \frac{1}{4} + \frac{B}{3}$$

$$\frac{1}{6} - \frac{1}{4} = \frac{B}{3}$$

$$-\frac{1}{4} = B \quad \text{--- (III)}$$

$$80 \left[\frac{1}{4s} - \frac{1}{4(s+2)} - \frac{2}{4(s+2)^2} \right]$$

$$20 \left[\frac{1}{s} - \frac{1}{(s+2)} - \frac{2}{(s+2)^2} \right]$$

$$20 \left[u(t) - e^{-2t} u(t) - 2t \cdot u(t) \cdot e^{-2t} \right]$$

$$20 \left[1 - e^{-2t} - 2t \cdot e^{-2t} \right] u(t)$$

Q5)

sdn

$$X(s) = \frac{2s+3}{(s+1)(s+3)}$$

$$= \frac{2s+4}{(s+1)(s+3)} - \frac{1}{(s+1)(s+3)}$$

$$= \frac{1}{s+1} + \frac{1}{s+3} - \frac{1}{2} \left[\frac{1}{s+1} - \frac{1}{s+3} \right]$$

$$X(s) = \frac{1}{2(s+1)} + \frac{3}{2} \frac{1}{(s+3)}$$

$$x(s) = \left[\frac{1}{2} e^{-t} + \frac{3}{2} e^{-3t} \right] \cdot u(t)$$

$$x(0^+) = \left[\frac{1}{2} + \frac{3}{2} \right]$$

$$= \frac{4}{2} = 2$$

Q6) $f(t) = f(2t) + u(2t)$

$$\mathcal{L}f(t) = \int_0^{\infty} f(2t) \cdot e^{-st} dt + \int_0^{\infty} u(2t) \cdot e^{-st} dt$$

$$\begin{aligned} 2t &= T \\ dt &= \frac{dT}{2} \end{aligned}$$

$$\int_0^{\infty} f(T) \cdot e^{-sT/2} \cdot \frac{dT}{2} + \int_0^{\infty} u(T) \cdot e^{-sT/2} \frac{dT}{2}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{2} \cdot \frac{\sqrt{2}}{5}$$

$$= \frac{1}{2} + \frac{1}{5}$$

$$= \frac{s+2}{2(s)}$$

Q.7) $y' - y = 1 \quad y(0) = 0$

$$\frac{dy}{dt} - y = 1$$

$$sF(s) - F(s) = \frac{1}{s}$$

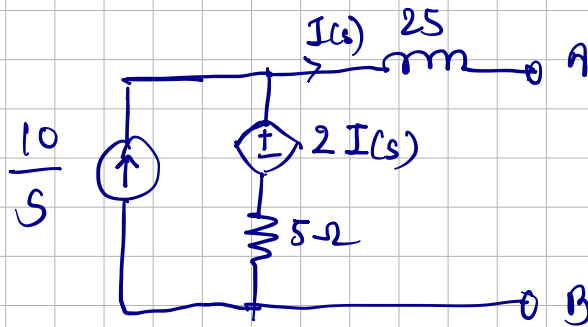
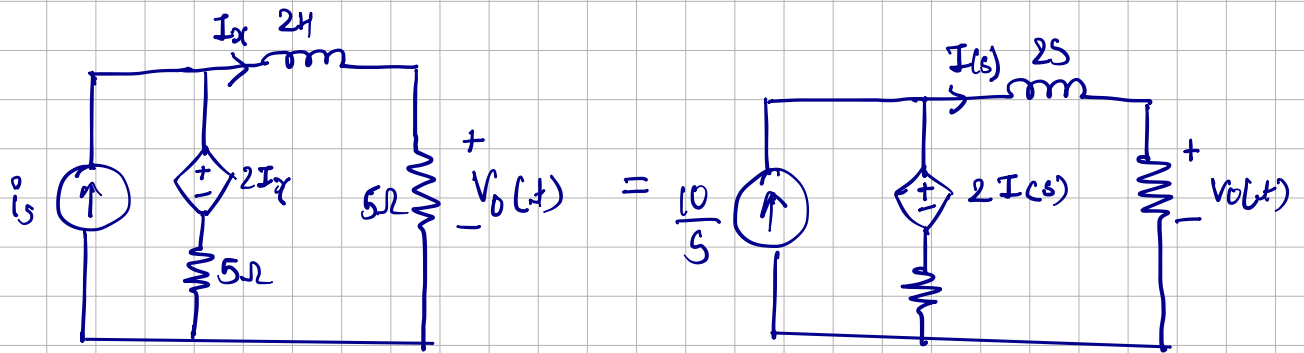
$$F(s) = \frac{1}{s(s-1)}$$

$$F(s) = \frac{1}{(s-1)} - \frac{1}{s}$$

$$F(t) = e^t u(t) - u(t)$$

$$y = (e^t - 1)$$

Q.8)

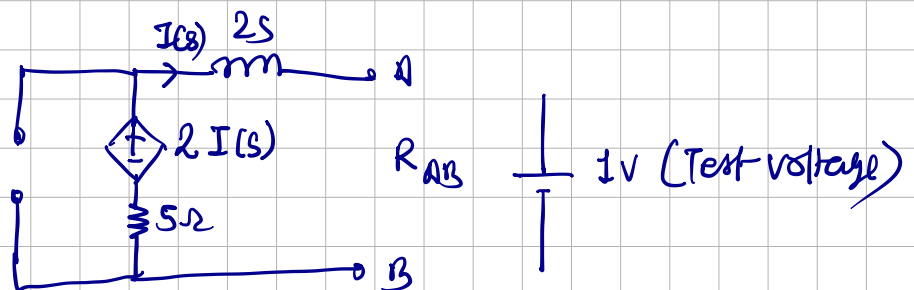


$$I(s) = 0$$

$$\therefore \frac{10}{s} \times 5 = V_{AB}$$

$$V_{AB} = \frac{50}{s} \quad \text{--- (1)}$$

now



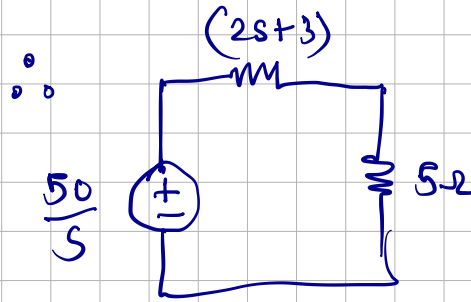
$$2I(s) - 2sI(s) - 1 - I(s) \cdot 5 = 0$$

$$I_s [2 - 2s - 5] = 1$$

$$I_s = \frac{1}{[-2s - 3]}$$

$$I_s = \frac{-1}{(2s + 3)}$$

$$\therefore R_{TH} = \frac{V}{I} = -(2s+3)$$



$$\therefore \left(\frac{\frac{50}{s}}{(2s+8)} \right) \cdot 5$$

$$= \frac{125}{s(s+4)}$$

Q9) $V_o(0^+)$ & $V_o(\infty)$

$$V_o(s) = \frac{125}{s(s+4)}$$

$$\begin{aligned} V_o(0^+) &= \lim_{s \rightarrow \infty} \frac{s \cdot 125}{s(s+4)} \\ &= \frac{125}{(s+4)} = 0 \end{aligned}$$

$$V_o(\infty) = \lim_{s \rightarrow 0} \frac{s \cdot 125}{s(s+4)} = 31.25 \text{ V}$$

$$V_0(s) = \frac{125}{s(s+4)}$$

$$= 125 \left(\frac{1}{4} \left(\frac{1}{s} - \frac{1}{s+4} \right) \right)$$

$$V_0(t) = \frac{125}{4} \left[u(t) - e^{-4t} \cdot u(t) \right]$$

$$= 31.25 \left(1 - e^{-4t} \right) u(t)$$